

Agentic AI - PDR

PRODUCT DESIGN REQUIREMENT (PDR)

AI-Powered Competitive Intelligence Platform

1. Product Overview

Product Name (Working): *IntelEdge AI*

Category: B2B SaaS (Competitive Intelligence Automation)

Vision:

Build an autonomous, multi-agent AI platform that continuously monitors competitors and delivers structured, actionable insights to businesses—without manual research.

Problem Statement:

Companies currently rely on scattered sources (news, LinkedIn, pricing pages, GitHub, etc.) to track competitors. This process is:

- Time-consuming
- Inconsistent
- Non-scalable

Solution:

An AI-powered system that:

- Tracks competitors automatically
 - Analyzes multiple data streams
 - Generates weekly intelligence reports
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2. Target Users

- Startups tracking competitors

- Product managers
 - Growth/marketing teams
 - Founders & analysts
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3. Core Features (Refined)

◆ Multi-Agent System

- Orchestrator Agent (controls workflow)
 - Specialist Agents:
 - News Agent
 - Social Media Agent
 - Pricing Tracker Agent
 - Job/LinkedIn Agent
 - GitHub/Product Updates Agent
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◆ Continuous Monitoring

- Scheduled scraping + API ingestion
 - Change detection (price updates, hiring spikes, etc.)
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◆ Intelligence Engine

- Summarization
 - Trend detection
 - Sentiment analysis
 - Competitor comparison
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◆ Output System

- Weekly reports (PDF + Email)

- Dashboard insights
 - Alerts (important changes)
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◆ Admin Dashboard

- Competitor management
 - Agent status monitoring
 - Logs + analytics
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4. Key Differentiator (Your Selling Point)

👉 Not just “AI summaries” — but:

“An autonomous competitive intelligence system with memory, trend tracking, and multi-source reasoning.”

5. System Architecture (High-Level)

🧠 Core Layers

1. Data Ingestion Layer

- Scrapers (Playwright)
- APIs (News, Twitter/X, etc.)
- Scheduled jobs (cron / workers)

2. Processing Layer

- Lightweight NLP models (local where possible)
- LLM (only when needed)
- Deduplication + filtering

3. Memory Layer

- Vector DB (Pinecone / Weaviate)

- PostgreSQL (structured data)
- Redis (cache)

4. Agent Layer

- LangGraph / CrewAI orchestration
- Stateful workflows

5. Output Layer

- FastAPI backend
 - Streamlit / React dashboard
 - Email service (SendGrid)
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6. Cost Optimization Strategy (VERY IMPORTANT)

You're right to worry about API costs. Here's the **smart approach instead of training full models**:

Use Hybrid Intelligence

1. Local Models (FREE / CHEAP)

- Use:
 - LLaMA / Mistral (via Ollama)
 - Sentence Transformers (embeddings)
 - Tasks:
 - Basic summarization
 - Classification
 - Similarity search
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2. Use LLM APIs ONLY for:

- Final report generation

- Complex reasoning
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3. Add Smart Caching

- Store:
 - Previous summaries
 - Competitor insights
 - Avoid re-calling APIs
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4. Batch Processing

- Process data in bulk instead of per request
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👉 Result:

You reduce API cost by **70–90% without training your own model**

7. Why NOT Train Your Own Model (for now)

Training your own model:

- ❌ Requires GPUs (₹₹₹)
- ❌ Needs large datasets
- ❌ Maintenance overhead

Better approach:

👉 Fine-tune later OR use open-source models locally

8. File Architecture (Clean & Scalable)

```
intel-edge-ai/
|
├── backend/
|   ├── api/           # FastAPI routes
|   ├── agents/       # All agents
|   │   ├── orchestrator.py
|   │   └── news_agent.py
```


- Backend → Railway / Render
- DB → Supabase / Neon
- Redis → Upstash
- Vector DB → Pinecone

Phase 3 (Production)

- Dockerized services
- CI/CD (GitHub Actions)
- Monitoring (LangSmith)

10. Pricing Strategy (Reality Check)

\$500/month is:

👉 Too high for early-stage product unless enterprise

Better Model:

Plan	Price	Features
Free	\$0	1 competitor
Starter	\$19–49	3 competitors
Growth	\$99–199	10 competitors
Pro	\$299+	Advanced analytics

👉 Later:

- Enterprise = \$500+

11. Resume / Pitch Line (Polished)

"Built a production-ready AI-powered competitive intelligence platform using a multi-agent architecture that processes 500+ daily data points, tracks competitor behavior across multiple sources, and generates automated intelligence reports with sub-3s response latency."

12. Next Step (Important)

If you're serious about making this *top-tier*, next we should:

👉 Design:

- **Detailed system architecture diagram**
- **Agent workflow (LangGraph flow)**
- **Database schema**
- **API endpoints**